Graphic Era HILL UNIVERSITY

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University under section 2(f) of UGC Act, 1956

# End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: B.Tech., III
Name of the Paper: Advanced Engineering Mathematics
Paper Code: BSC-301
Time: 3 hour

Maximum Marks: 100

**Note:**

*(i)* All the questions are compulsory.
*(ii)* Answer any two sub questions from a, b and c in each main question.
*(iii)* Total marks for each question is 20 (twenty).
*(iv)* Each sub-question carries 10 marks.

**Q1** (10 X 2=20 marks)

(a) Find complex Fourier integral representation for the function CO1

$$f(x) = \begin{cases} |x|, & if -\pi < x < \pi \\ 0 & otherwise \end{cases}$$

(b) Find the Fourier sine transform of $e^{-|x|}$ and hence evaluate $\int_0^\infty \frac{x \sin mx}{1+x^2}\, dx$.

(c) Define Z -transform and find the inverse Z-transform of $F(z) = \frac{z}{z^2-5z+6}$.

**Q2** (10 X 2=20 marks)

(a) If $f(z) = u + iv$ is analytic function and $u - v = e^x(\cos y - \sin y)$, find $f(z)$ in terms of z. CO2

(b) Show that the function $u(x, y) = \frac{1}{2}\log(x^2 + y^2)$ is harmonic and find its harmonic conjugates.

(c) Show that the function $f(z) = e^{-z^{-4}}; z \neq 0$ and $f(0) = 0$ is not analytic at z=0 although the C-R equations are satisfied at that point.

**Q3** (10 X 2=20 marks)

(a) If $f(z)$ is analytic function with in a closed contour with its center $z = a$ and radius r then at every point z inside C$f(z) = \frac{\sum_{n=0}^{\infty} f^n(a)(z-a)^n}{n!}$ CO3

(b) Evaluate by the method of calculus of residues:
$\oint \frac{dz}{(z-1)(z+1)}$, where C is circle $|z| = 3$.

(c) Find the kind of singularity of the function $f(z) = \frac{(z-2)}{z^2} \sin\left(\frac{1}{z-1}\right)$.

**Q4** (10 X 2=20 marks)

(a) Find the rank correlation coefficient for the following data: CO4

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| x= 68 | 64 | 75 | 50 | 64 | 80 | 75 | 40 | 55 | 64 |
| y= 62 | 58 | 68 | 45 | 81 | 60 | 68 | 48 | 50 | 70 |

(b) In a bolt factory, machine A, B and C manufactures 25%, 35% and 40% of the total of their inputs 5%, 4% and 2% are defective bolts. A bolt is drawn at random from the product and is found to be defective. What are the probabilities that it was manufactured by machine A, B or C ?

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(c) Define the following:

(i) Moment generating function

(ii) Normal distribution

(iii) Skewness and Kurtosis

**Q5** (10 X 2=20 marks)

(a) Fit the straight line to the following data: **CO5**

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| x= 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| y= 9 | 8 | 10 | 12 | 11 | 13 | 14 | 16 | 5 |

(b) Find the regression lines to the following data:

| | | | | | | |
|---|---|---|---|---|---|---|
| x= 5 | 7 | 8 | 10 | 11 | 13 | 16 |
| y= 33 | 30 | 28 | 20 | 18 | 16 | 9 |

(c) Fit a parabola $y = a + bx + cx^2$ to the following data:

| | | | | |
|---|---|---|---|---|
| x= 2 | 4 | 6 | 8 | 10 |
| y= 3.07 | 12.85 | 31.47 | 57.38 | 91.29 |